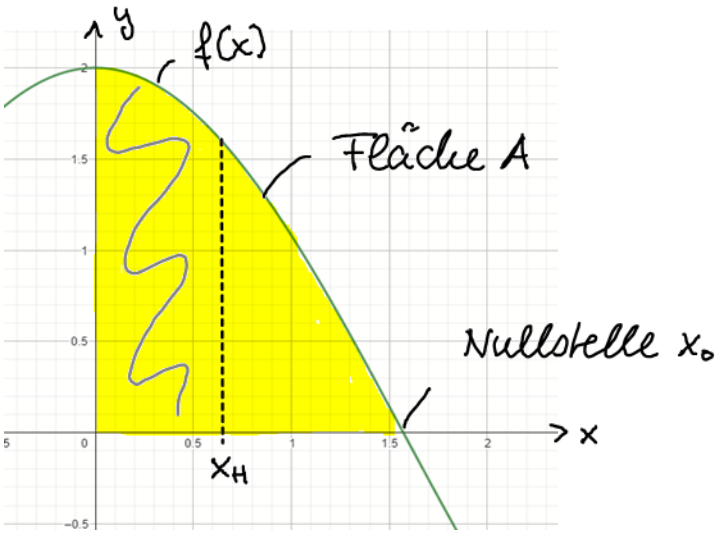
Lösung: Fläche zwischen Graphen



Nullstelle x_0:

$$f(x_0) \stackrel{!}{=} 0$$

$$2 \cdot \cos(x_0) = 0$$

$$\cos(x_0) = 0$$

$$x_0 = \arccos(0)$$

$$\underline{\underline{x_0 = \pi/2}}$$

$$A = \int_0^{\pi/2} f(x)\,dx = \int_0^{\pi/2} 2 \cdot \cos(x)\,dx = 2 \cdot \int_0^{\pi/2} \cos(x)\,dx$$

$$= 2 \cdot \left[\sin(x)\right]_0^{\pi/2} = 2\left(\sin\left(\tfrac{\pi}{2}\right) - \sin(0)\right) = 2\,(1-0)$$

$$\underline{\underline{A = 2}}$$

Halbe Fläche:

$$\int_0^{x_H} f(x)\,dx = \frac{A}{2} = 1$$

$$2 \cdot \int_0^{x_H} \cos(x)\,dx = 1$$

$$\left[\sin(x)\right]_0^{x_H} = 1/2$$

$$\sin(x_H) - \underbrace{\sin(0)}_{=0} = 1/2$$

$$\underline{\underline{x_H = \arcsin(1/2) \approx \frac{\pi}{6}}} \qquad (\approx 30°)$$